

MORGANS POINT, TX, USA

WMO: 997370

Lat: 29.682N

Lon: 94.985W

Elev: 10

StdP: 101.20

Time Zone: -6.00 (NAC)

Period: 05-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) 1.4	(c) 3.5	(d) N/A	(e) N/A	(f) N/A	(g) N/A	(h) N/A	(i) N/A	(j) 11.5	(k) 10.5	(l) 10.0	(m) 11.4	(n) 5.0	(o) 10	(p) 0.431

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 8	(b) 5.5	(c) 33.5	(d) N/A	(e) 32.8	(f) N/A	(g) 32.1	(h) N/A	(i) N/A	(j) N/A	(k) N/A	(l) N/A	(m) N/A	(n) N/A	(o) 3.7	(p) 150

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a) N/A	(b) N/A	(c) N/A	(d) N/A	(e) N/A	(f) N/A	(g) N/A	(h) N/A	(i) N/A	(j) N/A	(k) N/A	(l) N/A	(m) N/A	(n) N/A	(o) N/A	(p) N/A

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a) 11.2	(b) 9.2	(c) 8.0	(e) -0.9	(f) 35.8	(g) 2.1	(h) 1.1	(i) -2.4	(j) 36.6	(k) -3.6	(l) 37.2	(m) -4.7	(n) 37.8	(o) -6.2	(p) 38.6		
			DB													
			WB													

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec			
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)			
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	21.5	12.2	14.0	17.9	21.4	24.7	28.4	28.8	29.5	27.2	22.8	16.8	14.0	(6)	
(7)		DBStd	6.97	4.39	4.80	4.16	3.13	2.49	1.34	1.44	1.39	1.84	3.53	4.53	4.60	(7)	
(8)		HDD10.0	77	29	20	5	1	0	0	0	0	0	0	7	16	(8)	
(9)		HDD18.3	626	192	129	56	11	1	0	0	0	0	8	80	148	(9)	
(10)		CDD10.0	4279	97	132	249	343	456	552	583	604	517	398	210	139	(10)	
(11)		CDD18.3	1785	2	7	42	103	198	302	324	346	267	147	33	12	(11)	
(12)		CDH23.3	18117	1	7	41	341	1526	3651	4044	4496	2876	1057	61	17	(12)	
(13)		CDH26.7	6603	0	0	2	15	241	1431	1686	2104	942	182	2	1	(13)	
(14)	Wind	WSAvg	3.8	3.6	3.8	4.4	4.8	4.7	3.9	3.3	3.0	3.4	3.6	3.7	3.6	(14)	
(15)	Precipitation	PrecAvg	1486	126	87	96	91	132	142	137	123	151	144	116	119	(15)	
(16)		PrecMax	2056	297	197	241	237	210	399	350	270	369	451	286	222	(16)	
(17)		PrecMin	1004	14	24	25	13	3	15	16	29	27	6	15	51	(17)	
(18)		PrecStd	288	69	54	54	58	70	94	86	79	80	129	67	45	(18)	
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	22.9	24.4	26.1	28.4	31.0	34.3	34.0	35.0	32.9	30.8	26.6	25.3	(19)	
(20)			MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(20)
(21)		2%	DB	21.0	22.5	24.4	26.9	29.9	32.6	32.9	33.7	31.9	29.7	25.1	23.2	(21)	
(22)			MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(22)
(23)		5%	DB	19.7	21.1	23.5	26.1	28.9	31.8	32.2	33.0	31.1	28.6	23.9	21.7	(23)	
(24)			MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(24)
(25)		10%	DB	18.4	19.8	22.7	25.3	28.0	31.1	31.6	32.5	30.3	27.7	22.7	20.3	(25)	
(26)			MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(27)	
(28)			MADB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(28)
(29)		2%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(29)
(30)			MADB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(30)
(31)		5%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(31)
(32)			MADB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(32)
(33)		10%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(33)
(34)			MADB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(34)
(35)	Mean Daily Temperature Range	5% DB	MDBR	6.6	6.5	5.6	5.5	5.0	5.2	5.1	5.5	5.4	6.6	6.6	6.6	(35)	
(36)			MADB	7.8	7.4	5.2	5.4	5.6	6.0	5.7	6.2	5.9	6.2	6.6	7.8	(36)	
(37)			MCWBR	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(37)
(38)		5% WB	MADB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(38)
(39)			MCWBR	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.343	0.361	0.376	0.439	0.480	0.473	0.498	0.484	0.458	0.383	0.373	0.352	(40)		
(41)		taud	2.513	2.471	2.419	2.240	2.162	2.234	2.186	2.229	2.267	2.466	2.441	2.494	(41)		
(42)		Ebn at Noon	893	904	910	859	822	823	801	810	820	869	848	862	(42)		
(43)		Edh at Noon	89	101	114	140	152	141	148	140	131	101	95	88	(43)		
(44)	All-Sky Solar Radiation	RadAvg	2.89	3.26	4.48	5.36	6.00	6.22	5.99	5.73	4.97	4.36	3.30	2.56	(44)		
(45)		RadStd	0.27	0.55	0.38	0.38	0.42	0.48	0.55	0.47	0.52	0.53	0.34	0.30	(45)		

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46) Station Only		N/A	(e) N/A	(f) N/A	(g) N/A	(h) N/A	(i) N/A	(j) N/A	(k) N/A	(l) N/A	(m) N/A
(47) Regional (33 neighbors)		+0.34	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+113	+94

Nomenclature: See separate page